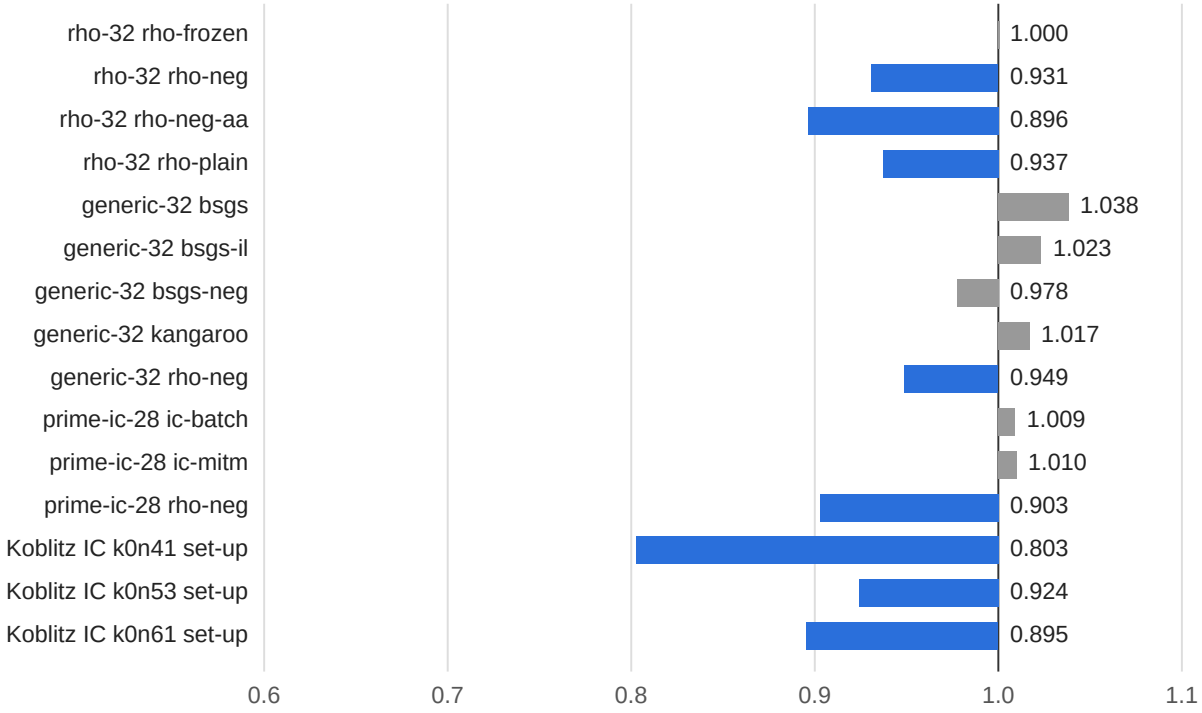
Algorithm performance, round 2 (2026-10-22)

Identical-output speedups in the Koblitz index calculus set-up and the prime-field Pollard rho walk. No counted quantity changes: 1776 of 1776 ecbench records are identical to the previous binary's outside timing, every session replays exactly under the other binary (`ecbench verify --replay-all`), the §5a band of the prime IC sessions is identical, and 0 `ic price` outputs differ in any count or answer.

New/old wall time per arm (below 1 = faster); counted results identical; grey = within ±5% (A/A noise)

ecbench: paired geometric mean of solve_wall_ns; Koblitz IC: median set-up of ic price (15 vs 15)



Koblitz index calculus set-up (`ic price`, rule-v3 T01, one thread, pinned)

curve	set-up old ms	new ms	ratio	phases ≥ 0.5 ms (old → new ms)	IC online ms	rho online ms
k0n41	57.2	45.9	0.803	collect 36.3 → 27.5, build 8.1 → 7.6, select 4.9 → 4.5, verify 2.5 → 2.5, collect_setup 2.1 → 0.7, select_projection 1.3 → 0.9, la 0.7 → 0.7, setup 0.6 → 0.6	1.21 → 1.19	3.0
k0n53	436.7	403.5	0.924	collect 234.8 → 211.5, build 134.5 → 132.3, select 23.6 → 23.2, collect_setup 11.5 → 4.2, verify 9.8 → 10.4, select_projection 7.1 → 5.3, descent_setup 1.8 → 1.0, la 1.6 → 1.6, setup 1.6 → 1.6, logs_setup 0.7 → 0.8	0.14 → 0.15	49.4
k0n61	1697.0	1519.6	0.895	collect 1386.8 → 1198.0, build 228.7 → 248.3, select 40.1 → 38.5, collect_setup 17.9 → 6.0, verify 13.8 → 14.3, select_projection 11.1 → 9.8, descent_setup 2.5 → 1.4, la 2.1 → 2.1, setup 1.9 → 2.2, logs_setup 1.2 → 1.2	7.68 → 7.80	28.5

Medians over 5 runs × 3 repetitions per binary. Changes: per-trial scratch buffers reused in the fast scan; a word add-mod in the walk; the shared index map and field structure built once per collector; Fx hash sets for the orbit factor base and the signed projection map, with the orbit stepped by k-fold squaring; aimed scans that pre-gather the target's doubled coordinates; the SIMD canonicaliser's tail handled as one padded block. Build and verify vary ±10% on unchanged code (noise).

ecbench sessions (prime-order curves)

spec	arm	method	old ms	new ms	median ratio	paired geo-mean	pairs
rho-32	rho-frozen	`rho.frozen_reference`	14.32	14.41	1.006	1.000	120
rho-32	rho-neg	`rho.negation`	9.98	9.34	0.936	0.931	120
rho-32	rho-neg-aa	`rho.negation`	10.21	9.46	0.927	0.896	120
rho-32	rho-plain	`rho.plain`	13.49	12.79	0.948	0.937	120
generic-32	bsgs	`bsgs.textbook`	14.96	14.81	0.990	1.038	36
generic-32	bsgs-il	`bsgs.interleaved`	15.67	15.83	1.010	1.023	36
generic-32	bsgs-neg	`bsgs.negation`	9.34	8.99	0.962	0.978	36
generic-32	kangaroo	`kangaroo.vow`	15.55	15.48	0.996	1.017	36
generic-32	rho-neg	`rho.negation`	10.85	10.23	0.943	0.949	36
prime-ic-28	ic-batch	`ic.pipeline`	61.97	62.03	1.001	1.009	18
prime-ic-28	ic-mitm	`ic.pipeline`	72.26	72.62	1.005	1.010	18
prime-ic-28	rho-neg	`rho.negation`	3.60	3.21	0.892	0.903	18

Changes in `ic_boundary`: the negation-map coefficient flip is a subtraction instead of two word divisions; inversion for 32-bit moduli runs Euclid on 32-bit words; the tuned walker computes each point's key and hash once per step. BSGS, kangaroo and prime IC do not take the changed paths enough to move (flat within ±4%).

Not done

Most of a prime rho step is its one field inversion (~110 ns at 32 bits). Batching inversions across walks (Montgomery's trick, in-order commit) would keep every walk and group-operation count but change the counted `field_ops` (inversions per step), which the accounting contract says must be what the code performed; that is an owner decision, not an identical-output change. Barrett multiplication and a quotient-1 Euclid shortcut were measured slower on this host (`microbench/`).